Urysohn's Lemma

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of urysohn's lemma, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

GT-03 / General Topology / Report v1

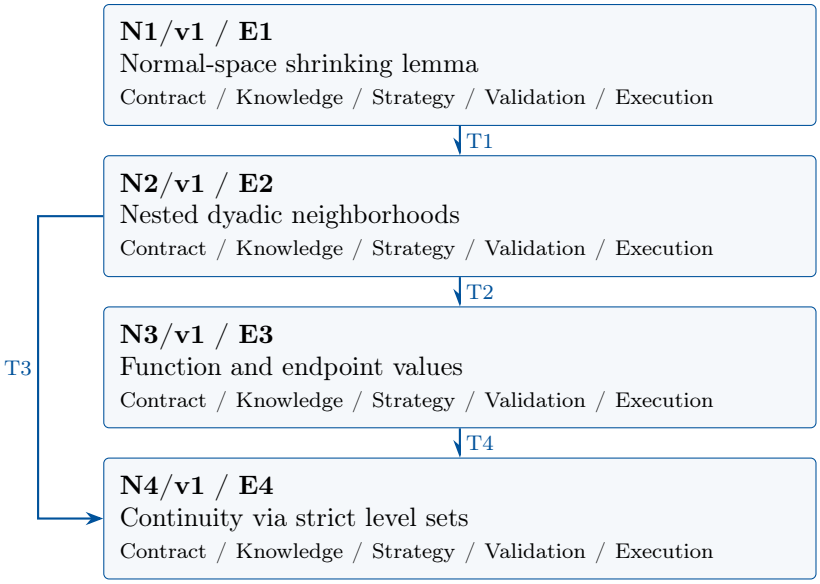
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For a normal T_1 space X and disjoint closed sets A, B, prove a continuous $f : X \to [0, 1]$ exists with $f = 0$ on A and $f = 1$ on B.*

A. Foundations and permitted imports

Normal means every pair of disjoint closed sets has disjoint open neighborhoods; T_1 means points are closed. Under this convention derive the shrinking lemma below. Dyadic rationals are dense in[0,1]. The construction uses closures in X and no metric. EmptyA or B is allowed.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. A neighborhood whose closure stays inside the prescribed open set.

Receiving activity. N2: Nested dyadic neighborhoods

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: InitialU $_0, U_1$ and shrinking lemma; prior exports E1/v1

T2: N2/E2/v1 to N3. A family with closure$(U_r) \subset U_s$ whenever $r < s$.

Receiving activity. N3: Function and endpoint values

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Nested neighborhoods; prior exports E2/v1

T3: N2/E2/v1 to N4. A family with closure$(U_r) \subset U_s$ whenever $r < s$.

Receiving activity. N4: Continuity via strict level sets

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Infimum definition and closure nesting; prior exports E2/v1, E3/v1

T4: N3/E3/v1 to N4. A function into[0,1] taking the specified endpoint values.

Receiving activity. N4: Continuity via strict level sets

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Infimum definition and closure nesting; prior exports E2/v1, E3/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Normal-space shrinking lemma

Object and role. Separate F from the closed complement.

Inputs. ClosedF inside open G; foundational imports stated in §1

C — Contract

Own E1/v1: A neighborhood whose closure stays inside the prescribed open set.

K — Knowledge

Retain ClosedF inside open G; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Separate F from the closed complement.

V — Validation

Disjoint open neighborhoods alone are insufficient unless the closure inclusion is obtained.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Normality gives disjoint open U,V containing F and $X \setminus G$.

N1.2 Since $U \subset X \setminus V$ and $X \setminus V$ is closed, closure$(U) \subset X \setminus V \subset G$.

N1.3 Thus $F \subset U \subset$ closure$(U) \subset G$.

N1.4 Take $G = X \setminus B$ and $F = A$ to choose U_0 with $A \subset U_0 \subset$ closure$(U_0) \subset U_1 = X \setminus B$.

Export E1/v1. A neighborhood whose closure stays inside the prescribed open set.

Acceptance gate. Disjoint open neighborhoods alone are insufficient unless the closure inclusion is obtained. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Disjoint open neighborhoods alone are insufficient unless the closure inclusion is obtained. Recheck dependent nodes N2, N3, N4.

B. N2/v1: Nested dyadic neighborhoods

Object and role. Insert midpoint neighborhoods between neighboring dyadics.

Inputs. InitialU $_0, U_1$ and shrinking lemma; prior exports E1/v1

C — Contract

Own E2/v1: A family with closure$(U_r) \subset U_s$ whenever $r < s$.

K — Knowledge

Retain InitialU $_0, U_1$ and shrinking lemma; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Insert midpoint neighborhoods between neighboring dyadics.

V — Validation

Bare $inclusions U_r \subset U_s$ do not supply the continuity argument.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 At each finite dyadic stage, for every consecutive $pair r < s$ already constructed choose $U_{r+s}/2$ using $F = \text{closure}(U_r) \subset G = U_s$.

N2.2 Thus closure$(U_r) \subset U_m id \subset$ closure$(U_m id) \subset U_s$.

N2.3 Induction defines U_r for all dyadic$r \in [0, 1]$, and for $any r < s$ their levels share a sufficiently fine stage, giving closure$(U_r) \subset U_s$.

N2.4 All U_r $with r < 1$ lie in $X \setminus B$.

Export E2/v1. A family with closure$(U_r) \subset U_s$ whenever $r < s$.

Acceptance gate. Bare $inclusions U_r \subset U_s$ do not supply the continuity argument. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Bare $inclusions U_r \subset U_s$ do not supply the continuity argument. Recheck dependent nodes N3, N4.

C. N3/v1: Function and endpoint values

Object and role. Encode membership by an infimum.

Inputs. Nested neighborhoods; prior exports E2/v1

C — Contract

Own E3/v1: A function into[0,1] taking the specified endpoint values.

K — Knowledge

Retain Nested neighborhoods; prior exports E2/v1; keep their assumptions and versions attached.

S — Strategy

Encode membership by an infimum.

V — Validation

The empty-infimum convention must be specified.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Let $f(x) = \inf\{r \in D : x \in U_r\}$, taking the value 1 if the set is empty, where D is the set of dyadics in[0,1].

N3.2 Then $0 \leq f \leq 1$.

N3.3 For $x \in A$, $x \in U_0$, hence $f = 0$.

N3.4 For $x \in B$ noU_r containsx, evenU $_1 = X \setminus B$, so $f = 1$.

Export E3/v1. A function into[0,1] taking the specified endpoint values.

Acceptance gate. The empty-infimum convention must be specified. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: The empty-infimum convention must be specified. Recheck dependent nodes N4.

D. N4/v1: Continuity via strict level sets

Object and role. Prove both types of strict level sets are open.

Inputs. Infimum definition and closure nesting; prior exports E2/v1, E3/v1

C — Contract

Own E4/v1: The required continuous separator.

K — Knowledge

Retain Infimum definition and closure nesting; prior exports E2/v1, E3/v1; keep their assumptions and versions attached.

S — Strategy

Prove both types of strict level sets are open.

V — Validation

Superlevel openness uses closures, not just open neighborhoods.

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 For $0 < a \leq 1$, $\{f < a\} = \cup_{r \in D, r<a} U_r$: membership on either side follows from the infimum and the existence of a dyadic membership level belowa.

N4.2 For $0 \leq a < 1$, $\{f > a\} = \cup_{r \in D, a<r<1}(X \setminus \mathrm{cl}(U_r))$.

N4.3 Ifx is outside closure(U_r), it belongs to noU_s with $s \leq r$, so $f(x) \geq r > a$.

N4.4 Conversely if $f(x) > a$ choose $dyadicsa < r < s < f(x)$, possible even when $f = 1$.

N4.5 Then $x \notin U_s$ and closure$(U_r) \subset U_s$, sox lies in that open complement.

N4.6 Level sets outside[0,1] are trivial.

N4.7 Both strict level types are open, proving continuity and Urysohn's lemma.

Export E4/v1. The required continuous separator.

Acceptance gate. Superlevel openness uses closures, not just open neighborhoods. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Superlevel openness uses closures, not just open neighborhoods. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Normal-space shrinking lemma

First let F be closed and G open with $F \subset G$.

1. Normality gives disjoint open U,V containing F and $X \setminus G$.
2. Since $U \subset X \setminus V$ and $X \setminus V$ is closed, closure$(U) \subset X \setminus V \subset G$.
3. Thus $F \subset U \subset$ closure$(U) \subset G$.
4. Take $G = X \setminus B$ and $F = A$ to choose U_0 with $A \subset U_0 \subset$ closure$(U_0) \subset U_1 = X \setminus B$.

B. §4.N2 — Nested dyadic neighborhoods

1. At each finite dyadic stage, for every consecutive $pairr < s$ already constructed choose $U_{r+s}/2$ using $F =$ closure$(U_r) \subset G = U_s$.
2. Thus closure$(U_r) \subset U_m id \subset$ closure$(U_m id) \subset U_s$.
3. Induction defines U_r for all dyadicr $\in [0,1]$, and for $anyr < s$ their levels share a sufficiently fine stage, giving closure$(U_r) \subset U_s$.
4. All U_r $withr < 1$ lie in $X \setminus B$.

C. §4.N3 — Function and endpoint values

1. Let $f(x) = \inf\{r \in D : x \in U_r\}$, taking the value 1 if the set is empty, where D is the set of dyadics in[0,1].
2. Then $0 \leq f \leq 1$.
3. For $x \in A$, $x \in U_0$, hence $f = 0$.
4. For $x \in B$ noU_r containsx, evenU $_1 = X \setminus B$, so $f = 1$.

D. §4.N4 — Continuity via strict level sets

1. For $0 < a \leq 1$, $\{f < a\} = \cup_{r \in D, r<a} U_r$: membership on either side follows from the infimum and the existence of a dyadic membership level belowa.
2. For $0 \leq a < 1$, $\{f > a\} = \cup_{r \in D, a<r<1}(X \setminus \mathrm{cl}(U_r))$.

3. Ifx is outside closure(U_r), it belongs to noU_s with $s \leq r$, so $f(x) \geq r > a$.
4. Conversely if $f(x) > a$ choose $dyadicsa < r < s < f(x)$, possible even when $f = 1$.
5. Then $x \notin U_s$ and closure(U_r) $\subset U_s$, sox lies in that open complement.
6. Level sets outside[0,1] are trivial.
7. Both strict level types are open, proving continuity and Urysohn's lemma.

□

V. WORKED ORCHESTRATION

In R take $A = \{0\}, B = \{1\}$; a concrete separator is $f(x) = \max(0, \min(1, x))$. The dyadic construction is needed for arbitrary normal spaces; it need not reproduce this particular function. If A is empty the constant 1 works, and if B is empty the constant 0 works.

A. Boundary and hypothesis audit

A candidate drops closure nesting and asserts an arbitrary membership-threshold function is continuous. Validation identifies that its superlevel sets may fail to be open. Repair the neighborhood construction before publishing the function; all later continuity claims depend on it.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.

5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

A candidate drops closure nesting and asserts an arbitrary membership-threshold function is continuous. Validation identifies that its superlevel sets may fail to be open. Repair the neighborhood construction before publishing the function; all later continuity claims depend on it.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- MIT Introduction to Topology: separation, Urysohn, and Tychonoff material.